



Research paper

Grid-connected hybrid renewable energy systems for supermarkets with electric vehicle charging platforms: Optimization and sensitivity analyses

A. Allouhi^{a,*}, S. Rehman^b^a Ecole Supérieure de Technologie de Fès, U.S.M.B.A, Route d'Imouzzar, BP 2427, Fez, Morocco^b Interdisciplinary Research Center for Renewable Energy and Power Systems Engineering (IRC-REPS), King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

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ABSTRACT

Coupling energy and transport sectors is critical for optimum utilization of renewable energy. The present work proposes a grid-connected photovoltaic/wind/battery hybrid power system to generate electricity for supermarkets. The supermarkets are located at three geographically distinct cities in Morocco and integrate electric vehicle (EV) charging stations in their parking areas. HOMER grid software is used for the optimal design of the proposed hybrid renewable energy system (HRES) with an accurate utility tariff corresponding to the Moroccan scenario. Dakhla, which is a high windy site, is found to be the best implementation site with maximum renewable energy fraction (REF), minimum COE and operating costs of 71.66%, 0.0841 \$/kWh, and 0.124 M\$/year, respectively. The optimal grid-connected system consisted of 107 kW_p of PV, 300 kW of wind, 12 batteries, and 65 kW capacity inverter for Dakhla site. The next identified best site is Casablanca while Fez remained at third place based on COE and annual operating costs. At all of these sites, the EV charging process is achieved, almost completely, with 7300 charging sessions annually. A series of sensitivity analyses is performed based on supermarket energy intensity, renewable energy availability, and carbon pricing and impact on the viability of HRES was evidenced.

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1. Introduction

Ever increasing global population, power demands, modern life styles, environmental pollution and at the same time growing awareness; people from all walks of life are pushing towards the utilization of clean and renewable sources of energy (Allouhi et al., 2022). Furthermore, new and innovative ideas and applications like water desalination (Lotfy et al., 2022), combined cooling/heating (Chakravarty et al., 2022), hydrogen production (Arsad et al., 2022) and EV (Costa et al., 2022) are being promoted to be implemented based on renewable energy sources. The EVs are being used in many countries and hence to facilitate the charging of these vehicles peer-to-peer charging stations are required at supermarkets, government official buildings, industrial premises, hospitals, schools, universities, recreational facilities, and so on. This study addresses the possibility of developing renewable energy based EV's charging stations at five supermarkets in Morocco. The forthcoming paragraphs are devoted to

discuss the EVs and electric vehicle charging platforms (EVCPs) and utilization of hybrid renewable energy systems (HRES) to supply power to supermarkets and EVCPs.

1.1. Sustainable development and net zero energy concept

The Sustainable Development Goals (SDGs) defined by all United Nations Member States seek to promote living conditions of people while protecting the environment, at the present time and in the future. There are 17 SDGs, which needs to be addressed urgently by all member countries (United Nations, 2022). The Division for DSDG in the United Nations Department of Economic and Social Affairs (UNDESA) sustains the SDGs and their related thematic problems, such as energy, water, climate, oceans, urbanization, transport, and science and technology. The 17 SDGs identified include among others no poverty, no hunger, healthy conditions and well-being, quality education, clean water and sanitation; affordable and sustainable energy as well as responsible consumption and production (United Nations, 2022). According to UNDESA, transport is one of the major sectors that need urgent attention to mitigate the atmospheric pollution, which is being addressed in the present work.

* Corresponding author.

E-mail addresses: allouhiamine@gmail.com (A. Allouhi), srehman@kfupm.edu.sa (S. Rehman).

Nomenclature

A_{pv}	Total area of the PV panels (m^2)
BAT_c	Rated battery capacity (Wh)
BAT_{pf}	Power flow in and out batteries (W)
CDC	Carbon Dioxide Cost (\$/tonne)
COE	Cost of electricity (\$/kWh)
CP	Contracted power (kVA)
DC	Direct current
DF	Derating factor (%)
E_{grid}	Energy from the grid (Wh)
E_{load}	Energy needed by the load (Wh)
E_{pv}	Theoretical energy generated in one hour (Wh)
$E_{pv,d}$	Theoretical energy generated during the day (Wh)
FMF	Fixed monthly fee (\$)
GEF	Grid Emission Factor (gCO_2/kWh)
GHI	Global horizontal irradiation (kWh/m^2)
I_{rad}	Solar radiation intensity (W/m^2)
$I_{rad-STC}$	Reference solar radiation (W/m^2)
ICI	Initial cash investment (\$)
NCF	Net cash flow (\$)
N_{inv}	Number of inverter units
NPC	Net present cost (\$)
P_c	Charging power (W)
P_d	Discharging power (W)
P_{inv}	Inverter output power (W)
P_{pv}	Theoretical power produced by PV system (W)
P_{pv-STC}	Power output of the PV field at standard test conditions (W)
P_{wt}	Wind power produced by wind turbine/s (W)
PD	Power demand (W)
REF	Renewable energy fraction (%)
PV_{ic}	PV installed capacity (W)
SCCP	Specific cost of contracted power (\$/kVA)
SOC	State of charge
T_{cell}	Current cell temperature ($^{\circ}C$)
$T_{cell-STC}$	Cell temperature standard test conditions ($^{\circ}C$)
T_h	Time interval during which certain incident radiation intensity persisted ($^{\circ}C$)
TAC	Total annualized cost of the system (\$/yr)
TES	Total energy served annually (kWh/yr)
WS	Wind speed (m/s)

Greek letters

β	Temperature coefficient of the PV panels ($\%/^{\circ}C$)
$\eta_{c/d}$	Battery charging/discharging efficiencies
η_{inv}	Inverter efficiency
η_{pv}	PV panels efficiency
σ	Hourly self-discharge rate

Abbreviations

AC	Alternating current
BEV	Battery electric vehicle
BSS	Battery storage system
CAGR	Compound annual growth rate
EREV	Extended-range electric vehicle
EV	Electric vehicle
EVC	Electric vehicle charging
EVCP	Electric vehicle charging platform
FCEV	Fuel-cell electric vehicle
FIT	Feed-in tariff
GHG	Greenhouse gases
HEV	Hybrid electric vehicle
HPS	Hybrid power system
HRES	Hybrid renewable energy systems
JEEP	Just Enough Essential Parts
PHEV	Plug-in hybrid electric vehicle
PV	Photovoltaic
RES	Renewable energy system
SDGs	Sustainable Development Goals
UNDESA	United Nations Department of Economic and Social Affairs

The international community has reached to a consensus that to avoid the worst climate damages, the global carbon dioxide (CO_2) and other greenhouse gas (GHG) emissions should reduce to net zero around 2050 and 2070. The Net-Zero Standard proposes a unique, strong, and science-based perception of net-zero concept. Net zero, simply, refers to the balance between the amount of GHG generated and the amount withdrawn from the atmosphere. Net zero scenario can be reached when the amount GHG added in to the atmosphere is almost equal to the amount taken away. Hence, in order to achieve net zero emissions, green and clean sources of energy have to be used in all sectors including transport, utilities, industries, residential, commercial, and public to name some.

1.2. Electric vehicle and electric vehicle charging platforms

In UK, almost 8% (2059) of the total electric vehicle (EV) charging units of 26,000 are located at supermarkets. In 2020, there were 1112 EV units located at supermarkets while in 2021 this number increased to 2059, an increase of 85%. The total number of superstores, having EV charging facilities has doubled from 607 in 2020 to 1300 in 2021 (Whitlock, 2021). From 2015 to 2021, the total sales of peer-to-peer electric vehicle charging market exceeded US\$ 800 Mn (FactMR, 2021). Furthermore, rapid adoption of EVs in developing countries and government incentives, EVs demand is projected to increase at a compound annual growth rate (CAGR) of 20% and will achieve US\$ 6.7 Bn between 2022 and 2032 (FactMR, 2021). The existing and future EV market is encouraging and hence should be promoted wherever possible. Various government have taken initiatives to promote the implementation of EVs.

EVs have started playing an important role in the public transportation system of many developed and developing countries. They are expected to have a positive impact (Boonraksa et al., 2018) such as reduction of greenhouse gases emissions into the atmosphere (Lata-García et al., 2018), high efficiency (Huang et al., 2014) and simple arrangement (Planas et al., 2015). Mainly, there are 5 types of commonly available EVs: (i) fuel-cell electric

vehicles (FCEVs), (ii) extended range electric vehicles (EREVs), (iii) hybrid electric vehicles (HEVs), (iv) plug-in hybrid electric vehicles (PHEVs), and (v) battery electric vehicles (BEVs) (CAA, 2022). The methods used to charge the EVs include (i) vehicle-to-vehicle (You and Yang, 2014), (ii) vehicle-to-grid (Contreras-Ocana et al., 2016), and (iii) vehicle-to-building (Contreras-Ocana et al., 2016). Mainly, the EVs are charged using Alternative Current (AC) or direct current (DC) and there are three standards for EVs which include the International Electrotechnical Commission (IEC), Society of Automotive Engineers (SAE), and CHAdeMO (CHAdEMO). EVs are driven by electric motors which are 80%–95% efficient (Canals et al., 2016) compared to conventional vehicles (CVs) which are 20% or less efficient (Åhman, 2001). This makes a perfect case of the promotion of EVs beside other many benefits. However, the growth of EVs, on global and regional levels, is still slow due to limited capacity of batteries which can be loaded on vehicle and slow charging process (Hardman et al., 2016; Kempton, 2016).

Some of the issues related to EV charging on the grid include voltage drop and fluctuations, grid overloading, harmonics, and system losses (Anoune et al., 2018; Karmaker, 2019) but its deployment can still bring a new look of power grid and ultimately EVs can be considered as portable energy storage media. Boonraksa et al. (2018) conducted a simulation study of a photovoltaic (PV) and wind hybrid power system (HPS) for EV charging station by modifying IEEE 33 bus system and concluded that the integration of the HPS with the distribution system proved to be reliable even during peak load times. In Morocco, a renewable energy system (RES) comprising of PV, wind, and grid; was proposed for charging EVs in transportation sector (Anoune et al., 2018).

A study of EVCP, based on PV-Grid HPS with battery storage system (BSS) in conjunction with efficient energy management strategy, was conducted by Eltoumi (2021). The proposed system resulted in improved overall performance, reliability, and the cost. Grid connected wind and solar PV based HPS with or without BSS are being considered to be attractive sources of energy for EVCPs as per the review analysis (Alkawsi et al., 2021). According to this survey, there exist a limited number EVCPs even for domestic EVs and hence new systems should be introduced for charging the heavy-duty vehicles. HOMER Pro software was used to study the techno-economic feasibility of a PV-Wind HPS for EVCPs for five locations in China and found that PV-Wind-Battery was the best option to serve the charging stations (Li et al., 2022). Furthermore, the most economically suitable system was reported for Nanjing and the least suitable for Zhengzhou in China. In Chile, an economic feasibility study of PV-Wind hybrid power systems, with and without battery, for supplying power to the residential buildings and providing electricity for EV charging station was conducted (Salles-mardones et al., 2022). The study found that the proposed HPS without battery backup was more profitable due to lower capital cost and shorter payback time.

To minimize the uncontrolled adverse effects of electric vehicle charging (EVC), a method was proposed for the hybrid EVCP coupled with PV and BSS by utilizing the exact calculation of renewable energy share coming from the battery (Fig. 1) (Petrusic and Janjic, 2020). The system consists of an energy source (PV) and energy storage (battery), and finally to the power grid. The EVs are charged with power available from renewable source and supplemented with battery and the grid if required. The extra power, at any time, is fed to the battery and then to the grid. The multi-criteria optimization methodology of charging/discharging schedule of a BSS and EVC level was based on the minimization of charging costs and battery degradation and maximization of renewable energy share. A PV powered EVCP, under varying irradiation conditions of Vietnam, was analysed and the authors reported that the economics is greatly dependent on solar irradiation intensity and the feed-in tariff (FIT) (Minh et al., 2021). Authors used HOMER Grid software to model the EVCP in combination with renewable sources of energy.

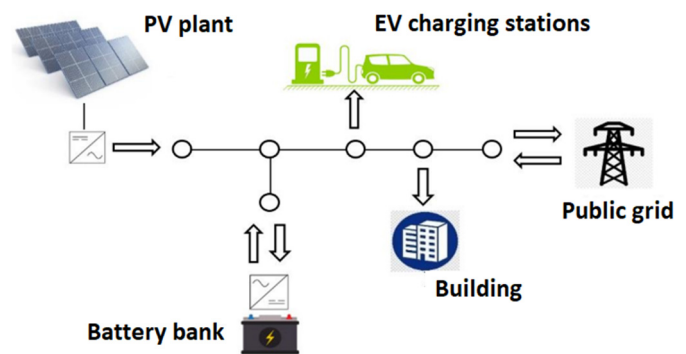


Fig. 1. PV-Battery electric vehicle charging platform (Petrusic and Janjic, 2020).

1.3. Renewable energy systems for supermarkets with electric vehicle charging platforms

Renewable energy systems have numerous applications and one such application providing power to superstores and at the same time feeding the EVCPs in parking areas to charge the EVs. The supermarkets are considered to be the highest energy consumers in the commercial sector (Franco et al., 2021). In the USA, the total number of Supermarkets and other grocery stores was about 63 thousand with sales value exceeding 765 billion \$ in 2021 (Statista, 2022). The energy intensity of supermarkets is around 537.6 kWh/m² which is more than 2 GWh of electricity annually for a normal supermarket (Sun et al., 2021).

According to Ong et al. (2013), the breakeven process of PV systems installed in supermarket across U.S. vary by a factor of >30. Deshmukh and Pearce (2021) investigated the possibility of developing PV canopies on parking areas of supermarkets and providing EVCPs for charging the EVs to the customers. The study revealed a potential of 3.1 MW for each Walmart Supercentre in the U.S. and this capacity is expected to charge more than 100 EVs. Further analysis showed that Walmart Supercentres in U.S. only could establish 11.1 GW installed capacity of PV canopies over their parking lots sufficient to charge >346,000 EVs.

It was reported that daily mean PV production potential of 4 kWh/kW in Italy suits to achieve a self-sufficiency of 50%–70% and hence the food stores will be the most suitable option for PV utilization either for self-consumption or for EVCPs (Franco et al., 2021). Usually, fast EV chargers are found at workplaces, supermarkets and shopping centres, train and bus stations, and airports (Cheikh-Mohamad et al., 2021). The Evergo company in Panama, completed the installation of 200 EV charging stations as of December 2021 and also established such EVCPs at Riba Smith superstores (Fig. 2), Costa del Este, Bella Vista, Transistmica, and Brisas del Golf (Angulo, 2020). PV based EVCPs have been installed at distribution centres in Basson and Centurion, 38 in supermarkets in South Africa and Namibia, and one in home office (Raghubir, 2021). Another example of EV charging station attached to a hypermarket is shown in Fig. 3.

1.4. Paper novelty and contribution

The current available literature has focused remarkably on the development of Hybrid Renewable Energy Systems (HRES) for several categories of buildings under both standalone and grid-connected applications (Zebra et al., 2021). Improving the energy efficiency of supermarkets with focus of critical applications such as refrigeration systems were investigated in early investigations such as Mota-Babiloni et al. (2022) and Tsimpoukis et al. (2021). Also, some works have dealt with the integration



Fig. 2. EVCP at Riba Smith superstore in Panama (Angulo, 2020).



Fig. 3. EVCP at Giant food (Redman, 2020).

of solar energy systems into supermarkets such as Franco and Cillari (2021) and Lykas et al. (2022). However, comprehensive analyses on supermarkets powered with HRES are not sufficiently investigated. The specifics of electricity load variation and energy intensity in supermarkets certainly has a great influence on the overall viability of HRES making their detailed analyses of extreme importance. Also, examining the available literature, it can be seen that most published works consider simplified billing assumptions to determine life cycle energy costs. These assumptions are sometimes not compatible with the real applied billing procedures. Supermarkets are medium-voltage electrical facilities, which are characterized by electricity costs that vary with respect to the different time bands. The present work analyses the feasibility of HRES in supermarkets considering this aspect respecting the Moroccan billing scheme in medium-voltage facilities. Furthermore, the present work proposes the design of an electricity charging station for small-size electric vehicles powered by the same HRES. The results from this investigation can help Moroccan authorities to have insight on the role that can play HRES systems to not only enhance the energy efficiency of supermarkets and lower their emissions but also how they can help in transitioning towards a clean and green mobility.

2. Methods

2.1. General description

The examined supermarkets have a total surface area of approximately 4200 m² consisting of one floor only. A typical supermarket electric load was imported from Homer Grid datasets. Integration of charging stations to charge small EVs is discussed in accordance with existing scenarios. The objective of the present analysis is to size an optimum HRES consisting of Solar PV/Wind/Battery system to fulfil sustainably the electric demand of the

Table 1

Geographical information of examined sites in Morocco.

City	Latitude (°)	Longitude (°)
Fez	34° 03' 00"N	4° 58' 59"W
Casablanca	33° 34' 44"N	7° 36' 23"W
Dakhla	23° 43' 00"N	15° 57' 00"W

supermarket and the EV charging station. HRES optimization is carried out for three climatically distinct cities in the Moroccan Kingdom, namely Fez, Casablanca, and Dakhla. The geographical information of these sites is given in Table 1. A general description of the followed methodology is displayed in Fig. 4.

2.2. HRES architecture

The proposed hybrid power system consists of wind turbines, power grid, and AC load of the supermarket and the EV charging station connected to an AC bus while the solar PV farm and the battery bank are connected to the DC bus with a converter in between the two buses, as depicted in Fig. 5. The PV farm and the wind turbines supply the power to the system and power grid, battery bank, and the EVCP work both way; means sometimes supply the power to the system and otherwise take the power from the HRES. Utility billing system of the available Moroccan grid for the target application and the mathematical background for each of the systems components are briefly provided in the forthcoming paragraphs.

2.3. Utility billing

To design appropriately the HRES, it is vital to describe accurately the electricity billing system since it significantly affects the overall project viability for grid-connected applications. The billing of supermarkets in Morocco relies on the medium voltage billing. The utilities install a transformer inside the supermarket and the monthly bills of electricity utilization depend on three periods of the day i.e. peak, normal, and off-peak hours. The energy consumption fee (CF) is calculated by taking the summation of the product of electricity price and consumption according to Eq. (1) (Boharb et al., 2016)

$$CF = \sum_{i=1}^3 EP_i EC_i \quad (1)$$

where EP_i and EC_i describe the electricity price and electricity consumption for the time period i , respectively. Additionally, a fixed monthly fee (FMF) is applied to account for the contracted power. Generally, the contracted power is identified through a consensus between the consumer and the electricity supplier utility. Specifically, it is determined based on forecasts of the maximum power demand that can occur in the future. This additional fee is billed on a monthly basis and is calculated using Eq. (2), as follows (Boharb et al., 2016)

$$FMF = SCCP \frac{CP}{12} \quad (2)$$

where $SCCP$ is the specific cost of contracted power (\$/kVA) and the CP is the contracted power (kVA). The electricity billing system as described by the Moroccan electricity utility is presented in Table 2.

2.4. Charging station

Since the EVs in Morocco are still a niche market, no sufficient data is available with respect to real operation of charging

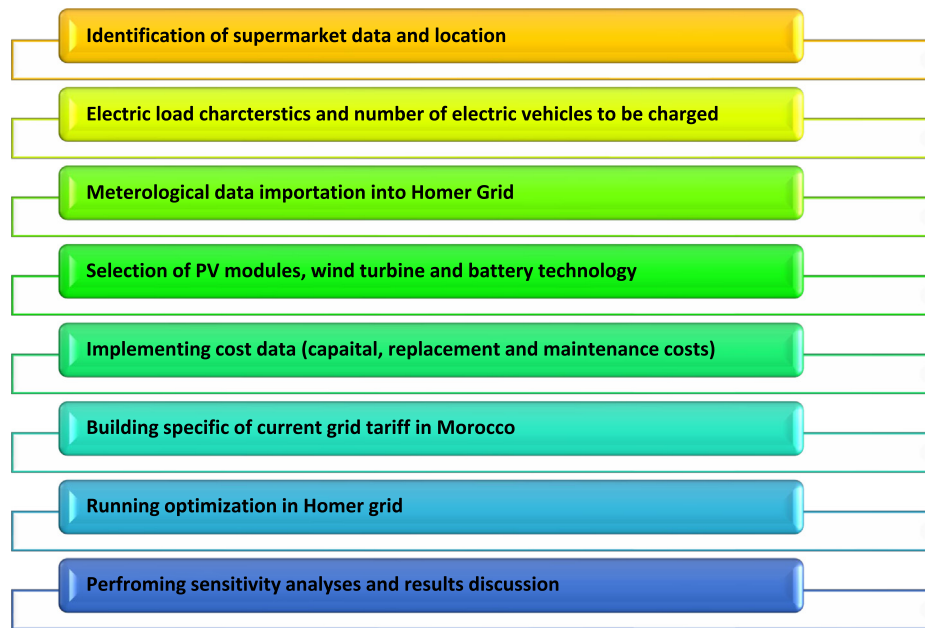


Fig. 4. Followed methodology in the present research.

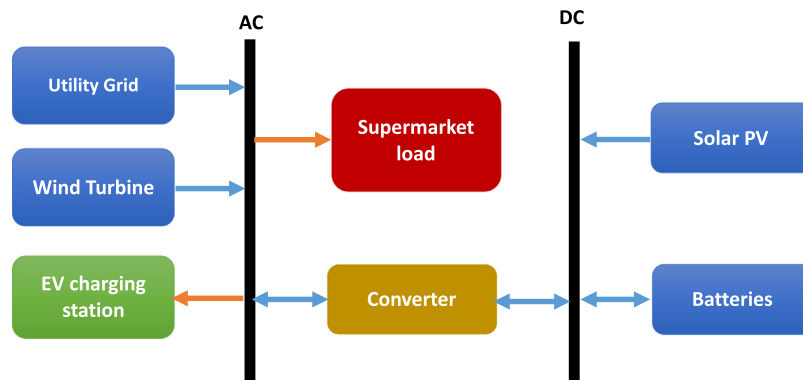


Fig. 5. A conceptual design of a PV-Wind-Battery-Grid hybrid power system with supermarket and EV charging platform.

Table 2
Electricity costs with corresponding time intervals.

	Off-peak hours	Normal hours	Peak hours
October–March	[22 h–07 h]	[07 h–17 h]	[17 h–22 h]
April–September	[23 h–07 h]	[07 h–18 h]	[18 h–23 h]
Electricity price (\$/kWh)	0.07	0.1	0.14
Specific cost of contracted power (\$/kVA)	51.2		

Table 3
Technical parameters of EV charging platform.

Parameter	Charger output power	Number of chargers	Average charging duration	Scaled average session per day
Value	50 kW	4	0.3 h	20

stations. A variable visit profile that changes from weekdays to weekends is adopted, as can be seen in Fig. 6. One can notice that higher number of customers to the supermarkets are observed during weekends compared to weekdays which reflects the overall behaviour of the supermarket occupancy. The technical characteristics of the charging stations constructed in the outdoor area of the supermarket are summarized in Table 3.

3. Mathematical description

This study uses Homer Grid as the optimization platform to appropriately design the HRES adapted for the supermarket case study. Homer Grid employs approximately the same mathematical background as Homer pro to compute energy generation of various renewable energy systems. It has additionally the ability

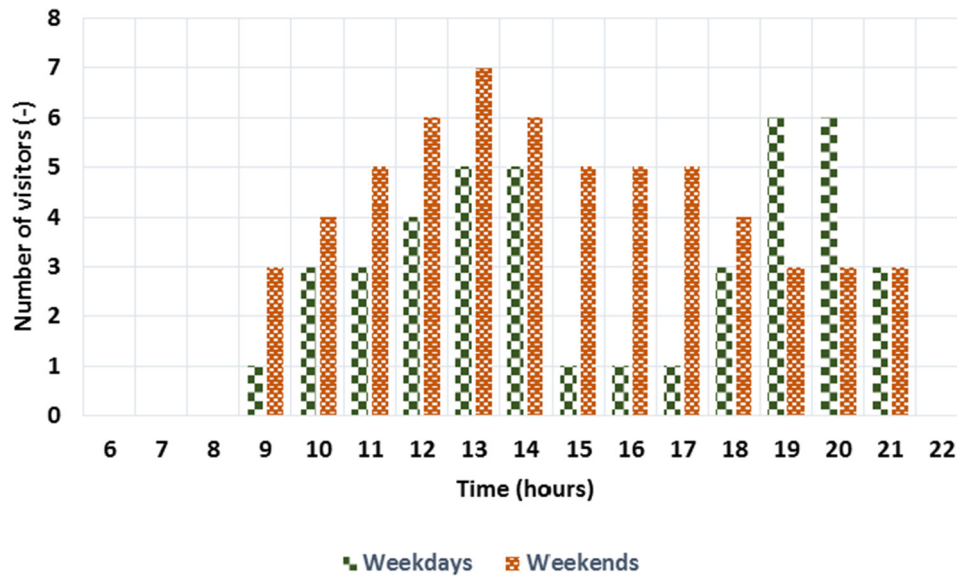


Fig. 6. Hourly profile of number of electric vehicles visiting the supermarket during weekdays and week-ends.

of introducing case-specific and more complex tariff structures adapted to grid-connected applications. Also, Homer Grid allows configuring EV charging stations and their coupling with hybrid renewable energy system. A brief mathematical description of computed outputs and optimization goals is furnished in the next subsections.

3.1. Solar PV

The PV output power P_{pv} from the arrays is calculated using the next formula:

$$P_{pv} = P_{pv-STC} DF \frac{I_{rad}}{I_{rad-STC}} [1 + \beta (T_{cell} - T_{cell-STC})] \quad (3)$$

where P_{pv-STC} , I_{rad} , $I_{rad-STC}$, T_{cell} and $T_{cell-STC}$ are the power output of the PV field at standard test conditions (W), DF is the derating factor (%), incident solar radiation (W/m^2), reference solar radiation ($=1000 W/m^2$), current cell temperature and cell temperature standard test conditions respectively. β denotes the temperature coefficient of the PV panels.

3.2. Wind turbines

The selected wind turbine is XANT M21 which has a rated power of 100 kW. It has the advantage of being mobile and can be constructed according to the JEEP (Just Enough Essential Parts) to lower the manufacturing costs. The technical specifications of the wind turbine are given in Table 4 and its power curve is displayed in Fig. 7. The solar PV power generation is simply simulated using a generic PV system of unit power of 1 kW_p. The tilt and the orientation angles are set equal to the regional latitude and 0° to maximize PV power output. Lithium-ion battery technology is selected as a storage system because of high efficiency, robustness, and cost reduction perspectives in the near future.

Based on available dataset for selected wind speeds and corresponded generated power, a curve fitting was applied. The power as function of wind speed can be then estimated as follows (Eq. (4)):

$$P = \begin{cases} 0 & \text{for } V \leq 3 \text{ m/s} \\ -0.03686V^4 + 0.9687V^3 - 7.603V^2 + 27.93V \\ \quad -36.68 & \text{for } 3 < V \leq 11 \text{ m/s} \\ 100 \text{ kW} & \text{for } 11 < V \leq 20 \text{ m/s} \\ 0 & \text{for } V \geq 20 \text{ m/s} \end{cases} \quad (4)$$

Table 4

Properties of the selected wind turbine XANT M21.

Characteristic	Value
Rated power	100 kW
Cut-in velocity	3 m/s
Rated velocity	11 m/s
Cut-out velocity	20 m/s
Hub height	31.8 m
Rotor Diameter	21 m

3.3. Battery storage

In the present study, the Modified Kinetic Battery Model is considered to simulate the battery storage system. The model integrates a series of resistance, temperature impacts on capacity and degradation rate. It includes as well the cycle-by-cycle degradation based on the depth of discharge. Datasheet from the manufacturer is used to define appropriately the mathematical model (Bhatt and Ongsakul, 2022)

3.4. Inverter dimensioning

Usually, the capacity of the inverters, which are used to convert the AC to DC and DC to AC, is kept slightly more than the installed PV capacity. This additional capacity allows inverter to operate always within its operating range and the losses due to overload are minimized. The power of the inverter can be calculated using Eq. (5) (Khan, 2019):

$$P_{inv} = \frac{PV_{ic}}{\eta_{inv} \cdot N_{inv}} \quad (5)$$

where P_{inv} is the inverter output power, PV_{ic} is the PV installed capacity, η_{inv} is the the inverter efficiency, and N_{inv} is the number of inverter units.

3.5. HRES economics

The cost assumptions made on the current case study are reassembled in Table 5. The feasibility study is examined based on two life cycle metrics, the total Net Present Cost (NPC) and the Cost of Electricity (COE). Homer grid is used to model the

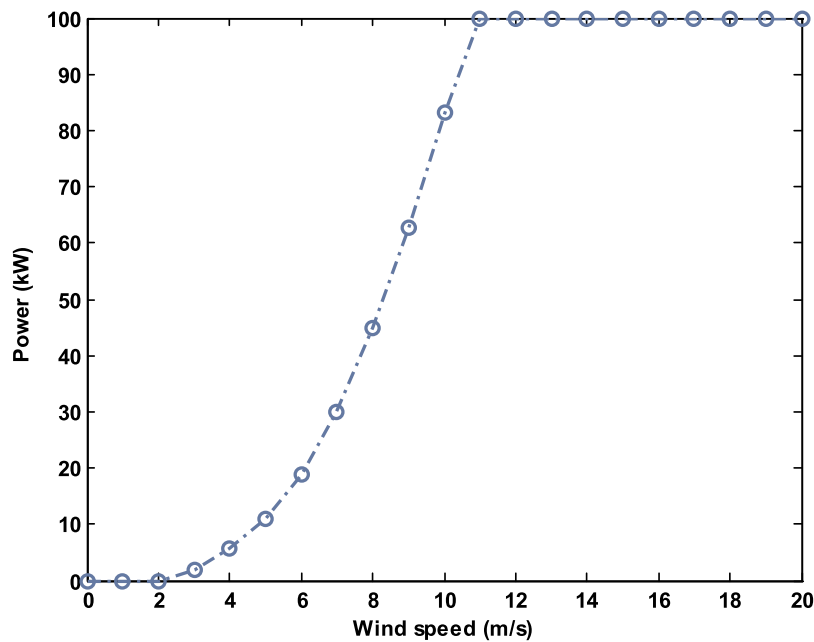


Fig. 7. Power curve of the wind turbine used.

Table 5

Cost consideration for the modelled HRES (Allouhi et al., 2021).

Component	Size	Capital cost (US\$)	Replacement cost (US\$)	O&M cost
Wind turbine	100 kW	250 000	250 000	62.5 [US\$/hr]
Battery	10.5 kWh	3500	2500	10 [US\$/yr]
Converter	1 kW	400	400	5 [US\$/yr]
PV	1 kW	1000	1000	10 [US\$/yr]

HRES and optimize its configuration and conduct the sensitivity analysis as well.

The total net present cost (NPC) of a HRES represents the difference between the present value of all costs needed over its lifespan and the present value of all the revenues it can generate during the same period. Capital, replacement and O&M costs are examples of needed expenses by the HRES. Financial gain resulting from grid sales and salvage are examples of revenues that can be generated by the HRES. Mathematically, the NPC is as follows (Kebede et al., 2021):

$$NPC = \frac{TAC}{CRF} \quad (6)$$

where TAC is the total annualized cost of the system while CRF is the capital recovery factor that can be assessed by:

$$CRF = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (7)$$

The Cost of Electricity (COE) is ratio between the total annualized cost and the total useful energy generation (TES) as shown by Eq. (8) (Bhatt and Ongsakul, 2022)

$$COE = \frac{TAC}{TES} \quad (8)$$

The Renewable Energy Fraction (REF) is the percentage of energy supplied to the supermarket coming from renewable energy calculated by the following expression:

$$REF = 1 - \frac{E_{grid}}{E_{load}} \quad (9)$$

where E_{grid} is the annual energy needed by the electric network to satisfy the unmet load by the HRES and E_{load} is the annual electricity demand of the supermarket.

4. Results and discussion

4.1. Weather data

Solar resources in terms of Global Horizontal Irradiance and clearness index are important when dealing the assessment and optimization of solar energy projects. The GHI represents the total solar radiation received by a horizontal surface while the clearness index is defined as the ratio between solar radiation on terrestrial surface and the extra-terrestrial radiation. These variables are illustrated on a monthly basis in Figs. 8 and 9. Monthly wind speed (WS) at the examined cities is depicted in Fig. 10. The chosen sites present a clear discrepancy in the prevailing weather conditions, which certainly will affect the overall design and viability of the proposed HRES.

4.2. Load profiles

The electric load variation of the supermarket is displayed in Fig. 11. This load profile corresponds to a typical Supermarket and was generated using the Homer Grid load online database based on real monitored data. For the present case study, the daily average electricity demand is 4806 kWh/day with peak value of 395 kW. The electricity demand of the charging station of small electric vehicles, based on the visitors' pattern, previously presented, is shown in Fig. 12.

The electric power demand of the charging station located in the outdoor supermarket surface presents a random profile reflecting the variable visitors' pattern. One can notice that for early morning and late-night hours, there is no electric energy demand. The maximum power demand of the charging station is 102 kW during certain hours of the day.

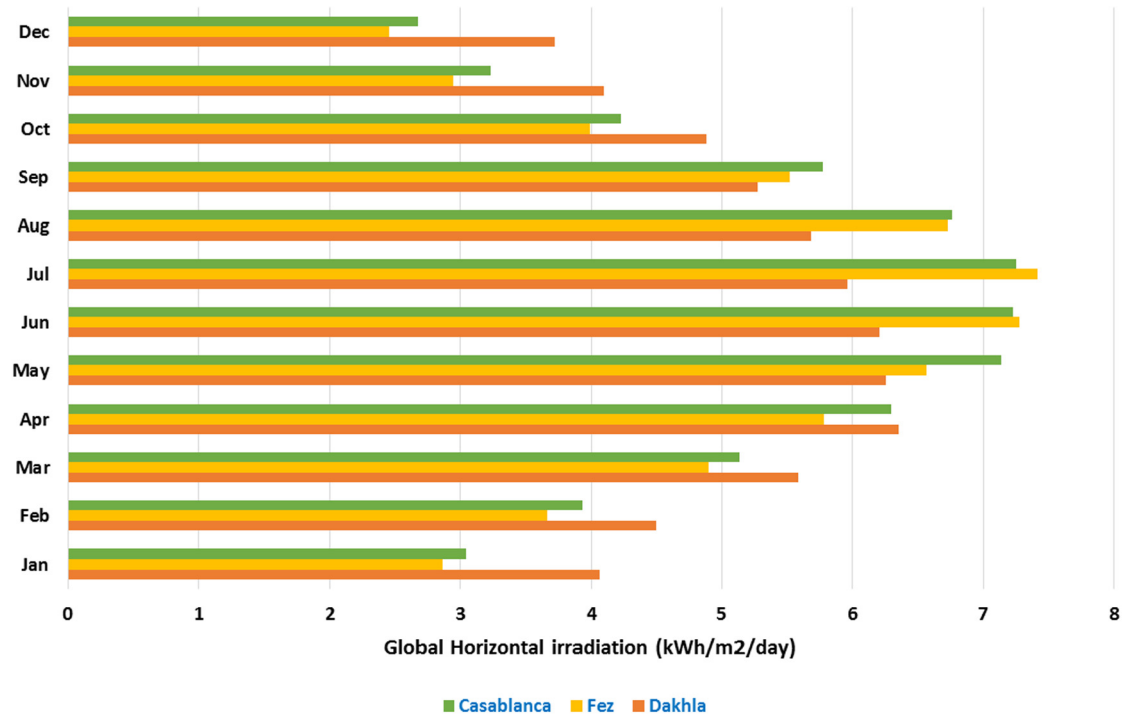


Fig. 8. Monthly variation of the GHI in the examined geographical sites.

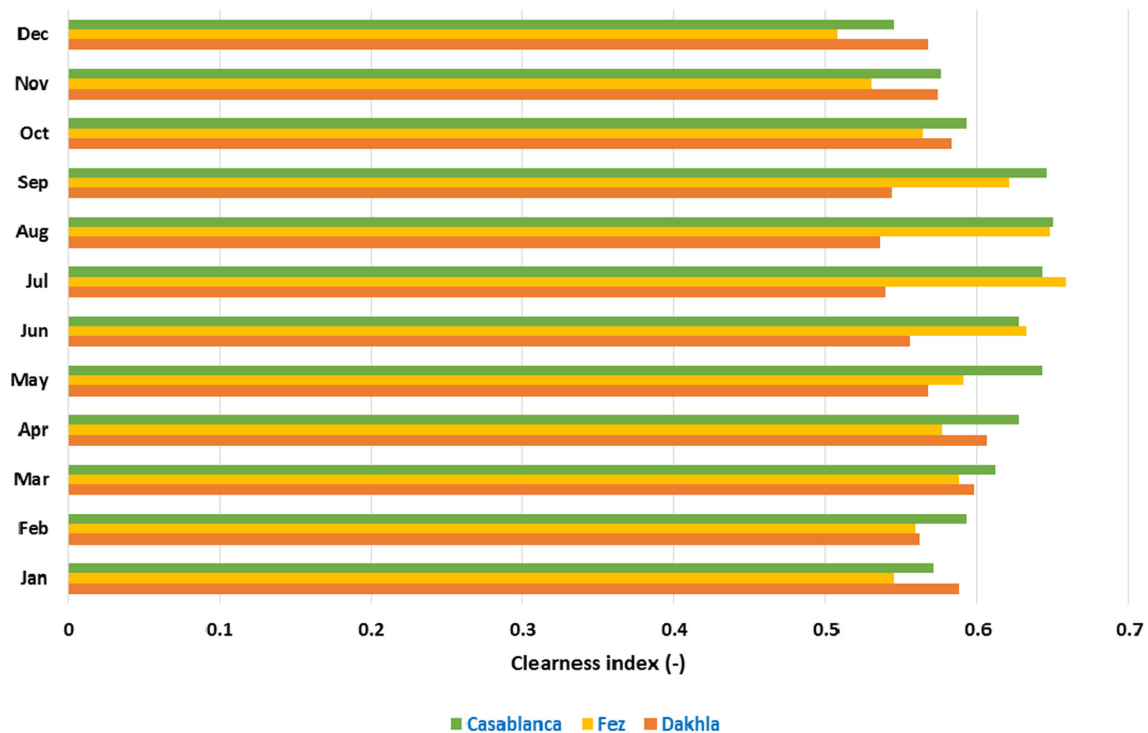


Fig. 9. Monthly variation of the clearness index in the examined geographical sites.

4.3. Optimization results

For each city, the best HRES configuration is obtained as indicated in Table 6. The designs are ranked from the best one (lowest NPC) to the worst one (highest NPC). The architecture, without renewable energy systems (baseline configuration), is summarized in Table 6 for the supermarkets located in Dakhla, Casablanca, and Fez as Design No. 7, Design No. 7, and Design

No. 4; respectively. The windy region Dakhla utilizes maximum number of wind turbines (3 corresponding to 300 kW installed capacity).

For the same city, the lowest PV capacity is recommended as 107 kWp with 12 EST1050 batteries and a 65 kW converter. The second recommended configuration has the lower share of renewable energy, which implies higher contribution of the utility grid (to be discussed later). The second case study at Casablanca,

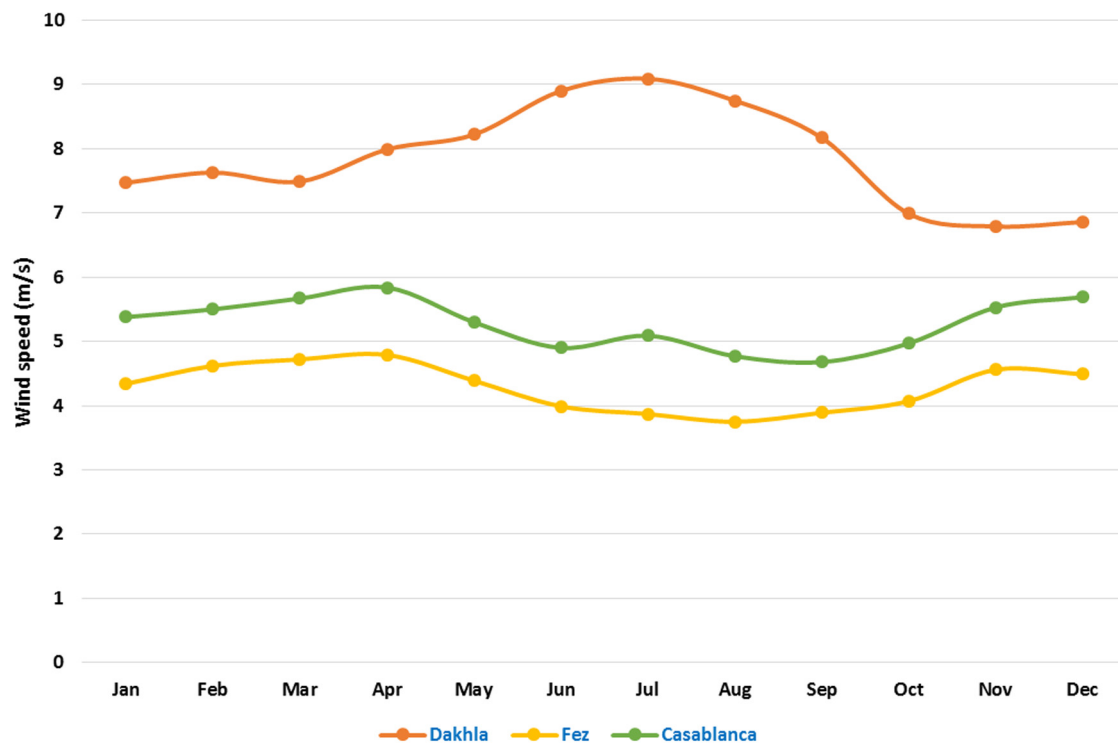


Fig. 10. Monthly variation of the WS in the examined geographical sites.

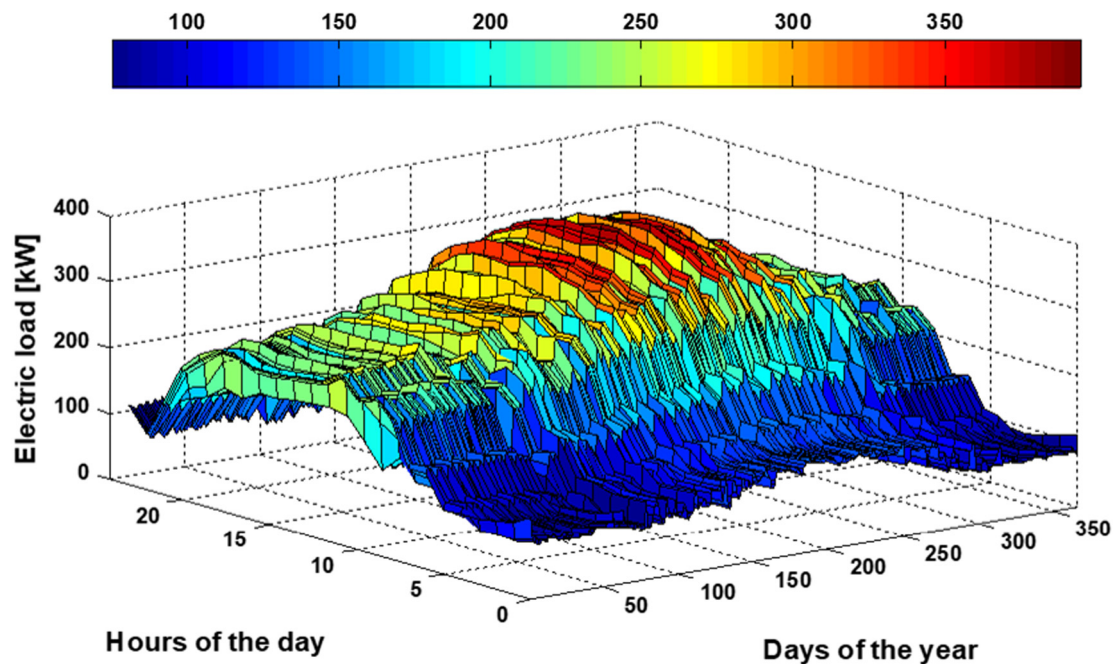


Fig. 11. 3-D representing the electric load in the studied supermarket.

utilizes higher PV capacity of 453 kW for the optimum Design No. 1 and a lower wind power capacity of 200 kW. In this case, larger energy storage (29 EST1050 batteries) is recommended compared to Dakhla and a larger converter size of 243 kW capacity. The city of Fez presents the highest contribution of solar PV power generation due to its relatively high GHI and lower WS values. The recommended PV size is 551 kWp connected to a converter of 325 kW. The optimal HRES for this site does not recommend the wind contribution and the battery storage.

For the same designs shown in Table 6 and for each one of the examined sites, the main performance metrics are displayed in Table 7. The most suitable sites for HRES with EVCP applied in supermarkets is Dakhla, for which the NPC and COE are 2.99 M\$ and 0.0841 \$/kWh, respectively for the optimal design. This site presents the highest REF (71.66%) and the lowest operating cost (0.124 M\$/year). It is evident that as the RE fraction increases, the operating cost decreases (Columns 5 and 7, Table 7). In fact, higher REF allows lesser electric energy import from the utility

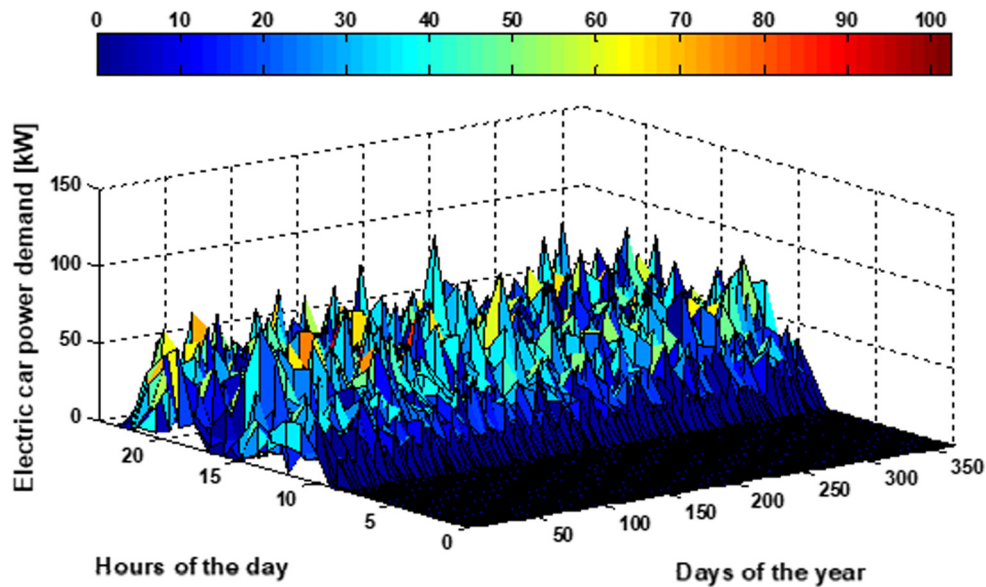


Fig. 12. 3-D plot representing the total electric power required for the charging station.

Table 6
Optimum HRES architectures in Dakhla, Casablanca, and Fez, Morocco.

City	Design No.	Solar PV (kW)	Wind turbine (XANT M21)	Battery storage (EST1050)	Grid	Converter (kW)
Dakhla	1	107	3	12	1	65
	2	63	3		1	41
	3		3		1	
	4		3	2	1	10
	5	568		2	1	318
	6	563			1	310
	7	Base case			1	
	8			2	1	5
Casablanca	1	453	2	29	1	243
	2	376	2		1	240
	3	560			1	325
	4	598		16	1	322
	5		3		1	
	6		3	3	1	6
	7	Base case			1	
	8			2	1	5
Fez	1	551			1	325
	2	572		4	1	326
	3	510	1		1	298
	4	Base case			1	
	5			2	1	5
	6		1		1	
	7	51	1	5	1	9
	8		1	8	1	14

grid which minimizes the yearly energy purchase expenditures. This observation is common for all the sites under consideration. The next best site for the implementing HRES is Casablanca with NPC and COE values of 3.55 M\$ and 0.1085 \$/kWh for the optimum HRES architecture. As in the previous case, the optimum design has the characteristic of having the largest REF of 60.45% and the lowest operating costs of 0.144 M\$/year. For Fez site, one can clearly see that the higher REF does not correspond to the best economic performance. Specifically, the highest REF (46.22%) was obtained in Design No. 3. The optimum Design No. 1 in Fez has lower REF of 41.85% and leads to NPC = 3.68 M\$, COE = 0.1153 \$/kWh, and operating costs = 0.18 M\$/year. Based on the optimization results of Tables 6 and 7 it can be seen also that unit cost of solar PV and wind electricity are respectively: 0.1147 and 0.0853 \$/kWh in Dakhla, 0.1143 and 0.1135 \$/kWh in Casablanca and 0.1153 and 0.1304 \$/kWh in Fez.

Fig. 13 illustrates the annual energy generation for optimum configurations for the three cities. The renewable energy contribution is the highest for Dakhla, with wind power share of about 64% and PV share of 8% of the total generation (including grid electricity). The supermarket located in Casablanca has a higher solar PV generation share of 37% compared to Dakhla. For this site, the PV generation is nearly equal to grid purchases while the remaining energy is obtained from the wind turbines (about 26%). As highlighted in Table 6, the supermarket located in Fez, wind energy system is not recommended. Hence, the total energy supply is ensured by PV system (44%) and the grid purchases (56%).

4.4. Sensitivity analyses

For the hybrid renewable energy systems discussed herein, wind speeds and solar incident solar radiation have a huge impact

Table 7
Performance indexes of various HRES in Dakhla, Casablanca and Fez, Morocco.

City	Design No.	NPC (M\$)	COE (\$/kWh)	Operating cost (M\$/yr)	Initial capital (k\$)	Renewable fraction (%)
Dakhla	1	2.99	0.0841	0.124	925.25	71.66
	2	3.01	0.0840	0.131	829.18	68.34
	3	3.02	0.0853	0.136	750.00	64.89
	4	3.02	0.0857	0.136	761.05	65.06
	5	3.66	0.1147	0.178	702.03	41.54
	6	3.66	0.1150	0.178	686.81	40.97
	7	4.06	0.1300	0.244	0.00	0.00
	8	4.06	0.1302	0.244	8.86	0.00
Casablanca	1	3.55	0.1085	0.144	1151.90	60.45
	2	3.56	0.1081	0.156	971.59	55.87
	3	3.63	0.1135	0.177	690.03	42.86
	4	3.63	0.1138	0.171	783.04	44.84
	5	3.74	0.1143	0.180	750.00	40.64
	6	3.74	0.1147	0.179	763.04	40.64
	7	4.06	0.1301	0.244	0.00	0.00
	8	4.06	0.1303	0.244	8.86	0.00
Fez	1	3.68	0.1153	0.180	680.99	40.88
	2	3.68	0.1153	0.178	716.19	41.85
	3	3.74	0.1167	0.172	879.14	46.22
	4	4.06	0.1300	0.244	0.00	0.00
	5	4.06	0.1302	0.244	8.86	0.00
	6	4.07	0.1304	0.230	250.00	8.96
	7	4.08	0.1307	0.226	321.78	10.95
	8	4.10	0.1312	0.229	283.60	8.76

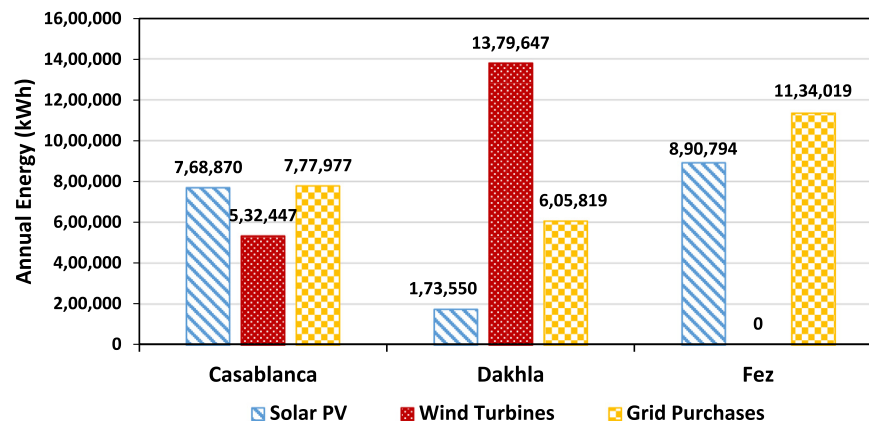


Fig. 13. Annual energy supply for optimum configuration in the three cities.

on the overall project viability. Their increase tends to maximize the energy output of wind turbines and PV panels. So, for the same design, variation if these climatic data affect the Net present Cost and Cost of Electricity of the HRES. Similarly, the optimization process is affected and the ranking of various HRES architectures will be changed.

The same supermarket load with charging station is used to conduct sensitivity analysis on the combined effect of varying annual global horizontal irradiance (GHI) from 4.8 to 6.8 kWh/m²/day and WS from 3 to 8 m/s, respectively. The HRES structure changes for each case resulting in different outputs such as NPC and COE values. Fig. 14 shows a contour representation of the NPC with respect to WS and GHI variation. Lowest values of renewable energy sources lead to the highest NPC indicating economically viable HRES projects. For the minimum and maximum values of both GHI and WS, the corresponding NPCs are 3.71 M\$ and 2.94 M\$, respectively. For this variation, the corresponding COE values are found to vary between 0.081 and 0.117 \$/kWh, for the best and worst cases, respectively. Here it can be said that as the WS and GHI values increase, the NPC decreases and vice versa. So, at lowest values of WS and GHI, the system becomes prohibitively unfeasible.

Annual electricity load has a significant effect on the size of HRES. Larger annual loads tend to increase the total capacity of

renewable energy system and the electricity imported from the public electric as well. These consequences have thus direct effect on the optimum design of the HRES and the NPC of the project.

To evidence the effect of electric load variation, the sensitivity of two scenarios is carried out corresponding to low and high energy intensity supermarkets. The scaled annual electric loads of 2403 kWh/year and 7210 kWh/year are considered for Dakhla with same EV charging platforms. The optimum architecture and economic indices as well as REF are summarized in Table 8. It can be seen that the NPC becomes more for high energy intensity supermarkets (3.97 M\$) with higher operating costs due to larger share from the grid compared to low energy intensity case. The low energy intensity scenario allows less installed capacity of wind without PV and battery storage for Dakhla climatic zone. It is seen that high intensity energy supermarkets are more competitive in terms of COE. This can be attributed to (i) higher RE generation from larger installed capacities to fulfil the annual electric load and (ii) the extra life-cycle costs are less influencing than the amount of energy produced for high electricity loads. The REF, which depends both on total renewable energy generation and electricity load does not exhibit significant change and remain close to 70%.

The effect of changing the number of electric vehicles served by the charging station was investigated by increasing gradually

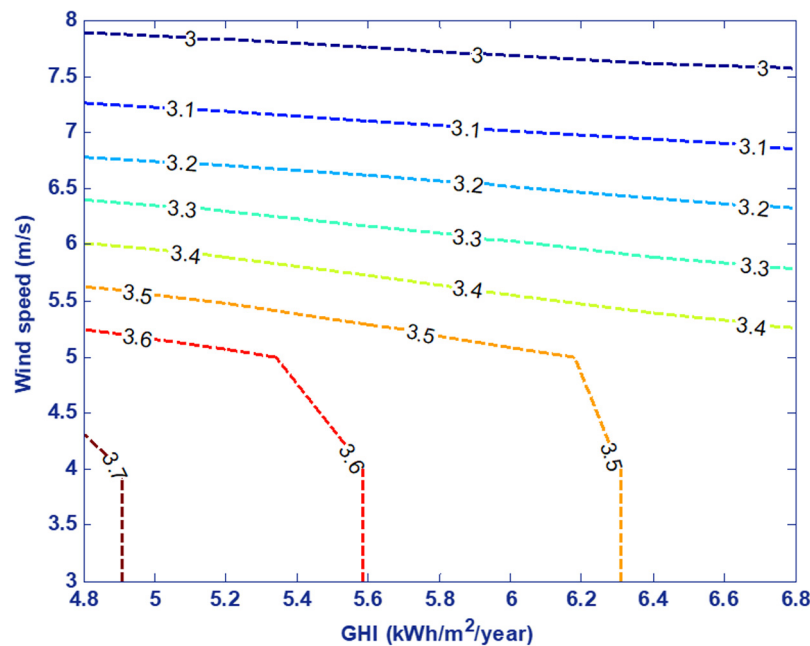


Fig. 14. Combined effect of GHI and WS on NPC (M\$).

Table 8

Optimum configurations and main performance outputs of the HRES in Dakhla according to two supermarket's energy intensities.

Scenario	Low energy intensity	High energy intensity
Solar PV capacity (kW)	–	221
Wind Turbine (XANT M21)	2	4
Battery storage (EST1050)	–	20
Grid	1	1
Converter capacity (kW)	–	129
NPC (M\$)	2.03	3.97
COE (\$/kWh)	0.097	0.078
Operating costs (M\$/year)	0.092	0.158
REF (%)	72.96	70.58

the number of scaled daily charging sessions from 10 to 60. The sensitivity analysis considers the climate conditions of Dakhla and optimization results are summarized in Table 9. It can be clearly seen that NPC of the HRES tends to increase principally because larger solar PV capacities and battery storage are needed to compensate for the increased energy requirements. The solar PV capacity increased from 90 kW_p for the lower number of sessions to 229 kW_p for the highest one. The renewable energy fraction and Cost of Electricity (COE) seem to be less sensitive to the number of charging sessions per day.

The environmental sensitivity analysis, describing how carbon dioxide emissions penalties influence the overall HRES project viability is examined. Optimization considers the combined effect of carbon dioxide cost penalty (ranging from 30 to 90 \$/ton) and the carbon dioxide density of grid electricity (ranging from 300 to 600 gCO₂/kWh). These variation intervals are realistic for both the current CO₂ market and for the Moroccan grid condition as highlighted in Allouhi (2022) (see Fig. 15).

Since renewable energy systems such as wind turbines and solar PV panels allow for a clean energy generation, they have the ability to reduce substantial quantities of CO₂ emissions compared to conventional ways such as fossil fuel combustion. Larger the emission factor of the public grid means that the energy mix of the country is greatly dependent on fossil fuels. Associating carbon dioxide costs in the financial analysis generally improves the feasibility of renewable energy systems.

Table 10 shows how the architecture of the HRES and its performance indices change with respect to grid emission factor (GEF) and carbon dioxide cost (CDC). Dakhla city in Morocco is considered for the sensitivity analysis with same supermarket electric load and charging station platform. It can be seen that inclusion of the pricing cost along with grid emission factor allows an increase of the renewable energy fraction from 74.4% to 84% with increased solar PV and storage capacities. This resulted in low operating costs by limiting energy purchases from the grid due to CO₂ penalties. However, the NPC remained higher while incrementing the carbon dioxide prices and the grid emission factor, principally due to high initial investment cost of renewable energy. So, it can be said that the investment cost proved to be more impactful than the reduction of operating costs.

5. Concluding remarks

The transport sector is an important aspect of renewable energy utilization widely, effectively and efficiently. The present study conducted a simulation to identify optimal HRES (PV/wind/battery/grid) integrated with EV charging platforms at supermarkets at three cities in Morocco. The study used the load profile, WS, solar radiation, and technical specifications of the major components of the HRES as input to the HOMER grid software and obtained best system for the supermarket in Dakhla, the second best for Casablanca, and the last one for Fez. Dakhla, being the windiest and hence the best choice whereas economics is concerned. Fez, being the least windy and is the worst case in economic terms. Some of the highlights of the present study are as follows:

- Higher values of GHI are observed during summer time with highest in July and the lowest in January at almost all the sites. The clearness index varied between 0.55 and 0.65, irrespective of the location factor.
- Highest WS values are observed in Dakhla with maximum in July and minimum in November. At Casablanca and Fez, highest wind speeds are observed in April while the lowest in August but much less in magnitude compared to Dakhla site.

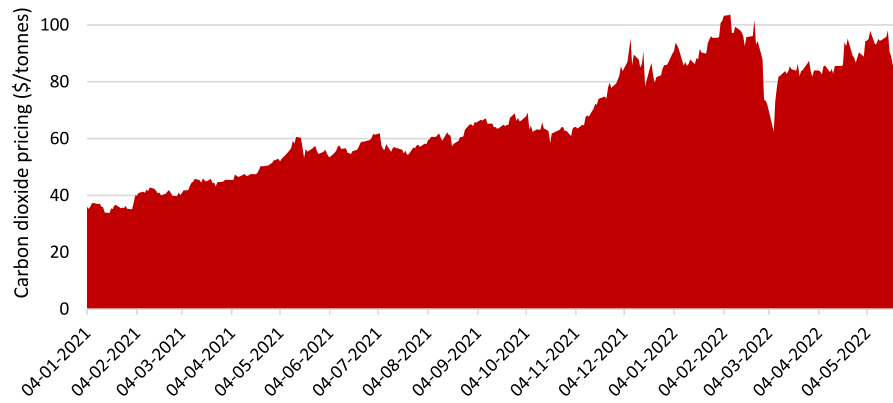


Fig. 15. History of carbon dioxide cost during 2021–2022 (Ember-Climate, 2022).

Table 9

Optimum configurations and main performance outputs of the HRES in Dakhla with respect to various daily sessions of charging.

Daily sessions	Solar PV (kW)	Wind turbine (XANT M21)	Battery storage (EST1050)	Grid	Converter (kW)	NPC (M\$)	COE (\$/kWh)	Operating cost (M\$/yr)	Renewable fraction (%)
10	90	3	11	1	59	2.92	0.0838	0.121	72.17
20	107	3	12	1	65	2.99	0.0841	0.124	71.66
30	134	3	14	1	81	3.07	0.0843	0.126	71.77
40	163	3	19	1	99	3.14	0.0846	0.127	72.27
50	195	3	22	1	111	3.23	0.0849	0.130	72.20
60	229	3	24	1	127	3.31	0.0850	0.132	72.53

Table 10

Combined effect of carbon dioxide pricing and grid emission factor on the HRES optimization.

CDC (\$/tonnes)	GEF (gCO ₂ per kWh)	Solar PV (kW)	Wind turbine (XANT M21)	Battery storage (EST1050)	Grid	Converter (kW)	NPC (M\$)	COE (\$/kWh)	Operating cost (M\$/yr)	Initial investment (M\$)	Renewable fraction (%)
30	300	156	3	17	1	96	3.08	0.0859	0.1247	1.0037	74.4
	400	171	3	22	1	103	3.10	0.0867	0.1242	1.0389	75.4
	500	180	3	22	1	111	3.13	0.0872	0.125	1.0512	75.8
	600	188	3	25	1	113	3.16	0.0880	0.1254	1.0705	76.3
60	300	188	3	25	1	113	3.16	0.0880	0.1254	1.0705	76.3
	400	223	3	27	1	129	3.21	0.0886	0.1254	1.1196	77.9
	500	229	3	29	1	132	3.25	0.0899	0.1274	1.1333	78.3
	600	268	3	36	1	153	3.30	0.0906	0.1258	1.2047	80.2
90	300	225	3	29	1	132	3.23	0.0894	0.1262	1.1291	78.1
	400	268	3	36	1	153	3.30	0.0906	0.1258	1.2047	80.2
	500	299	3	38	1	168	3.36	0.0915	0.1269	1.2489	81.5
	600	345	3	52	1	197	3.42	0.0924	0.1238	1.3561	84.0

- The wind power contributed the maximum share of renewable energy at Dhakla (1379.65 MWh/year) while solar PV contributed more at Casablanca (768.87 MWh/year) and Fez (890.79 MWh/year). Maximum power was purchased from the grid at Fez (1134.02 MWh/year) while the minimum at Dakhla (605.79 MWh/year).
- In general, the COE and annual operating costs are found to be decreasing with increasing RE fraction at almost all the sites.
- Highest NPC, COE, and annual operating costs are observed for base case (zero renewable energy fraction) at all the sites.
- The electricity requirements for powering EV were completely satisfied for a yearly number of charging sessions of about 7300 at all the sites.
- The optimization process of the HRES for the supermarkets revealed the best economic viability is obtained in Dakhla region with a NPC of 2.99 M\$, a COE of 0.0841 (\$/kWh) and REF of 71.66%.
- Sensitivity analysis, for several input parameters was carried out and corresponding modifications of optimum HRES architecture were studied. Supermarket energy intensity and

renewable energy sources potential showed pronounced effects on the HRES design and economics.

Based on the present analysis, the following recommendations can be highlighted:

- Additional feasibility studies assisted by Multi-Criteria Decision-Making techniques are crucial for real implementation of HRES with EV charging stations in Supermarkets
- Regulatory authorities can play a vital role by encouraging and incentivizing the supermarket owners to develop renewable energy based EV charging stations. The effect of incentive scenarios on the economics of the HRES are thus very important.
- HRES with Combined Power and Heat outputs in other utilities such hospitals can as well have huge energy and environmental benefits for more sustainable cities in Morocco.

CRedit authorship contribution statement

A. Allouhi: Conceptualization, Methodology, Software, Visualization, Investigation, Writing. **S. Rehman:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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